

AutoAnalog-RL: Reinforcement-Learning-Driven Device Sizing for a PCIe Gen 2 Receiver Equalizer in a 130 nm Open PDK

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Abstract: Device sizing for a high-speed receiver equalizer is a constrained search over continuous device values in which every specification is coupled through the same transistor dimensions, and in which the governing figure of merit, the eye opening after a lossy channel, is observable only from a transient simulation. Exhaustive parameter sweeps scale exponentially with the number of sized devices and are therefore impractical at the resolution required for a converged design. This paper presents AutoAnalog-RL, a sizing framework in which a Soft Actor-Critic agent interacts directly with ngspice and learns a sizing policy from simulated outcomes alone. The design target is a source-degenerated continuous-time linear equalizer with a one-tap decision-feedback stage for a PCIe Gen 2 (5 Gbit s⁻¹) receiver, implemented in the IHP SG13G2 130 nm open process design kit using foundry PSP103 compact models. Each environment step executes a fail-fast sequence of DC, AC, transient-eye and linearity gates, so that infeasible candidates are rejected at the earliest analysis that can reject them. The trained policy reaches a design satisfying every target specification, across all 45 process-voltage-temperature corners, in a median of 2.53 simulations from an arbitrary starting point, against 9 for CMA-ES and 13 for random search; measured against an exhaustive sweep at practical grid resolution, this is a reduction in sizing time of roughly four orders of magnitude per design instance. A two-stage corner-randomised curriculum yields a policy that remains feasible at corners withheld during nominal training (24/24 rollouts), a natural-language interface maps engineering intent onto reward weights, and each run emits the sized netlist as its deliverable. The optimisation additionally surfaced two latent defects in the measurement pipeline that manual sweeps had not exposed; both are analysed, corrected and covered by regression tests, and are reported here as a methodological result.

1 Introduction

Serial links at PCIe Gen 2 rates and above operate over channels whose insertion loss rises steeply with frequency. At the 2.5 GHz Nyquist frequency of a 5 Gbit s⁻¹ NRZ link, a representative FR-4 trace attenuates by roughly 10 dB, and the resulting inter-symbol interference closes the received eye entirely. Receiver-side equalization restores it: a continuous-time linear equalizer (CTLE) applies a high-frequency boost that inverts the channel's loss slope, and a decision-feedback equalizer (DFE) cancels residual post-cursor interference without amplifying noise [2, 3].

Sizing such a front end is the difficulty. The five device values of a source-degenerated CTLE (input width, load resistance, tail current, degeneration resistance and degeneration capacitance) set gain, bandwidth, power, linearity and noise jointly, so no specification can be closed in isolation. The specification that ultimately governs link operation, the vertical and horizontal eye opening after the channel, is not visible in a small-signal analysis at all; it requires a pseudo-random bit sequence transient through an explicit channel model [1]. An exhaustive sweep over five continuous parameters is exponential in the grid resolution: even a coarse ten-point grid is 10⁵ simulations, which at the throughput achieved here is roughly 17 h of wall-clock time for a single design instance, and a twenty-point grid is over three weeks. General-purpose optimisers reduce this count but must repeat the entire search whenever the target specification, the process corner or the relative weighting of objectives changes.

Reinforcement learning addresses precisely this repetition. A policy trained once over a distribution of starting points and operating conditions amortises its cost: sizing a new instance becomes a short sequence of policy evaluations rather than a fresh search. This idea has been advanced for analog sizing generally, and system-level surveys of machine-learning-assisted SerDes design identify simulation-in-the-loop optimisation of the receiver front end as a principal application [4]. The framework presented here realises it for a complete PCIe Gen 2 equalizer on foundry-grade device models.

1.1 Contributions

1. **A closed simulation-in-the-loop sizing framework.** A Soft Actor-Critic agent drives ngspice directly over a bounded five-parameter action space, with no surrogate model and no human intervention between the target specification and the sized netlist. Candidate evaluation is organised as a fail-fast gate cascade (DC $\rightarrow$ AC $\rightarrow$ transient eye $\rightarrow$ linearity), so an infeasible design is rejected at the cheapest analysis capable of rejecting it (Section 3).
2. **Sizing on foundry compact models, not idealised devices.** All reported results use the IHP SG13G2 PSP103 surface-potential models, compiled from Verilog-A to OSDI and executed natively in ngspice (Section 3.3). A thread-pinning correction to the simulator configuration yields a $65\times$ throughput improvement under parallel execution, which is what makes training on a laptop-class machine tractable.
3. **Convergence in a median of 2.53 simulations, quantified against baselines.** Deterministic rollouts reach a fully feasible design in 2–3 simulations across 144 rollouts, against 9 for CMA-ES and 13 for random search from equivalent starting points, and against an estimated 10^5 evaluations for an exhaustive sweep at comparable resolution (Sections 6 and 6.3).
4. **Reward shaping that produces margin rather than marginal compliance.** Terminating an episode at the first feasible design leaves the policy on the constraint boundary. Holding the episode open and paying an explicit margin bonus moves the median eye height from 0.43 V to 0.92 V against a 0.5 V target (Section 4.2).
5. **Corner robustness by domain randomisation.** A second training stage in which each episode is simulated at a corner drawn uniformly from the 45-corner matrix, with the corner identity withheld from the observation, yields a policy that reaches feasibility at randomly drawn corners in 24 of 24 rollouts (Section 7).
6. **Specification retargeting without redesign.** The Nyquist frequency is a framework input that propagates to the measurement frequency, data rate, unit interval, transient step and channel pole. Resuming the trained policy at 1.25 GHz recovers full compliance within 3k steps, demonstrating the tunability required of the peaking stage (Section 7.3).
7. **Measurement validation as a by-product of optimisation.** Two defects in the measurement pipeline—spectral leakage in the HD3 estimator and a decision-labelled DFE eye—were surfaced by the optimisation reaching regions of the design space that manual sweeps had not visited. Both are analysed, corrected, and covered by regression tests (Section ??).

2 Design Target

2.1 Specifications

Table 1 states the target specifications, the analysis that measures each, and the internal optimisation target used by the reward. Where the two differ, the internal target is deliberately tighter: optimising against the acceptance threshold produces designs that sit on it, whereas optimising against a tightened target produces designs that retain margin against it under corner variation. The eye targets are the clearest case—the acceptance threshold is 100 mV and 0.4 UI, while the reward requires 0.25 V and 0.7 UI.

2.2 Circuit topology

The equalizer is a differential NMOS pair with resistive loads and an RC source-degeneration network (Figure 1), followed by a one-tap decision-feedback stage. Degeneration introduces a zero at $1/(2\pi R_s C_s)$ and a pole at higher frequency, so the stage attenuates low-frequency content relative to its mid-band gain and produces the high-frequency boost that inverts the channel response. Five values are sized: input device width W_{in} , load resistance R_{load} , tail current I_{bias} , degeneration resistance R_s and degeneration capacitance C_s . All devices are IHP sg13_1v_nmos at the minimum drawn length $L = 130$ nm; the bounds in Table 2 define the action space.

2.3 Channel and decision-feedback model

The transient gate drives a 127-bit PRBS7 sequence $(x^7 + x^6 + 1)$ at 5 Gbits^{-1} , repeated twice so that the measured period is free of start-up transients, through a two-section RC channel. Each section contributes a real pole at 1.7 GHz, giving approximately -10 dB of loss at the 2.5 GHz Nyquist frequency and -20 dB at 5 GHz—the loss slope characteristic of a PCIe Gen 2 class FR-4 trace. The DC and AC gates bypass the channel, so the reported peaking figure characterises the CTLE alone and remains directly comparable to the specification.

Specification	Target	Optimisation target	Measured by
Nyquist frequency	2.5 GHz (PCIe Gen 2, 5 Gbit s ⁻¹)	framework input, tunable	.ac / .tran
High-frequency peaking	3–12 dB, tunable over 1.25–2.5 GHz	identical, enforced in reward	.ac, 10–10 000 MHz
Eye height after channel	≥ 100 mV	≥ 0.25 V	PRBS7 .tran
Eye width after channel	≥ 0.4 UI	≥ 0.7 UI	PRBS7 .tran
Linearity, HD3 at 100 MHz	≤ -30 dB	identical, enforced with --hd3	single-tone .tran + FFT
Input-referred noise	≤ 1.5 mV rms, 10–5000 MHz	reported at validation	.noise
Power	≤ 15 mW	≤ 2 mW, plus continuous charge	.op
Active area	≤ 0.05 mm ²	reported at validation	device geometry
PVT coverage	TT/SS/FF/SF/FS, $V_{DD} \pm 5\%$, 0–125 °C	corner-randomised curriculum	45-corner sweep

Table 1: Target specifications, the tightened internal targets used by the optimisation, and the analysis that measures each. The 2 mW power target is set where the 2 mA tail-current bound actually binds (2.4 mW), rather than at the far looser 15 mW acceptance limit.

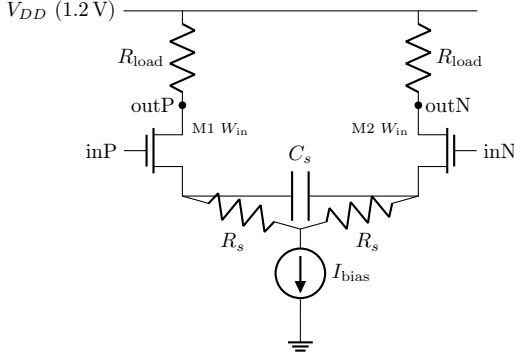


Figure 1: Source-degenerated differential CTLE. The degeneration network (R_s, C_s) sets the peaking zero; the five sized values are labelled.

Parameter	Lower	Upper
W_{in}	0.5 μ m	50 μ m
R_{load}	100 Ω	1000 Ω
I_{bias}	0.1 mA	2 mA
R_s	10 Ω	500 Ω
C_s	1 fF	1 pF
DFE tap b_1 (optional)	-0.5	+0.5

Table 2: Bounded action space. Normalised actions in $[-1, 1]$ map linearly onto these ranges; the sixth value is present only in the whole-equalizer configuration.

The decision-feedback stage (Figure 2) operates on the unit-interval centre samples of the equalized waveform and implements

$$y[n] = x[n] - b_1 \hat{d}[n-1], \quad (1)$$

where $\hat{d}[n-1]$ is the slicer’s own previous decision, so that error propagation is represented rather than assumed away. The tap b_1 is either swept over its bounds during validation or, in the whole-equalizer configuration of Section 7.2, sized by the policy as a sixth action.

3 Sizing Framework

3.1 Architecture

Figure 3 shows the closed loop. The policy emits a normalised design vector; the evaluator maps it onto device values, renders a parameterised netlist, and invokes the analysis gates in ascending order of cost. The resulting metrics are reduced to a scalar reward and returned, together with the design vector and the normalised violation of each constraint, as the next observation. Only the simulator adapter is technology-specific: the specification, reward and environment layers operate on metric dictionaries and are independent of both the simulator and the process.

3.2 Fail-fast evaluation

The four analyses differ in cost by more than an order of magnitude: an operating point resolves in milliseconds, an AC sweep in tens of milliseconds, and a two-period PRBS transient on PSP103 models in approximately 1.5 s.

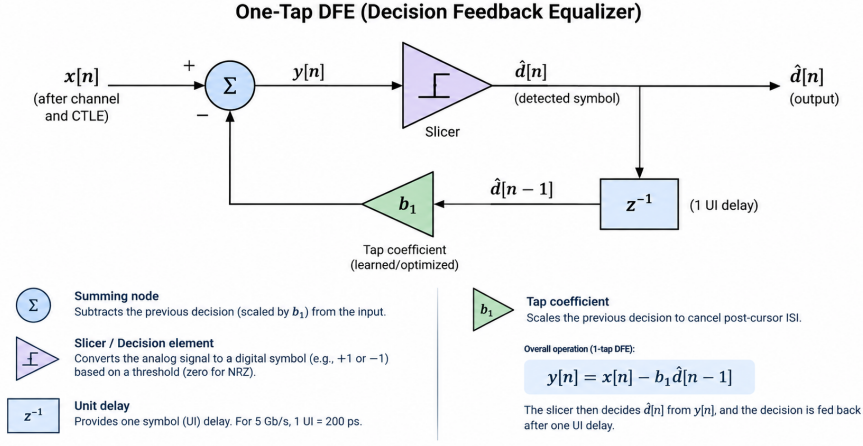


Figure 2: One-tap decision-feedback equalizer. The previous slicer decision, scaled by the tap coefficient b_1 and delayed by one unit interval (200 ps at 5 Gbit s^{-1}), is subtracted from the incoming sample to cancel post-cursor interference, per Equation (1).

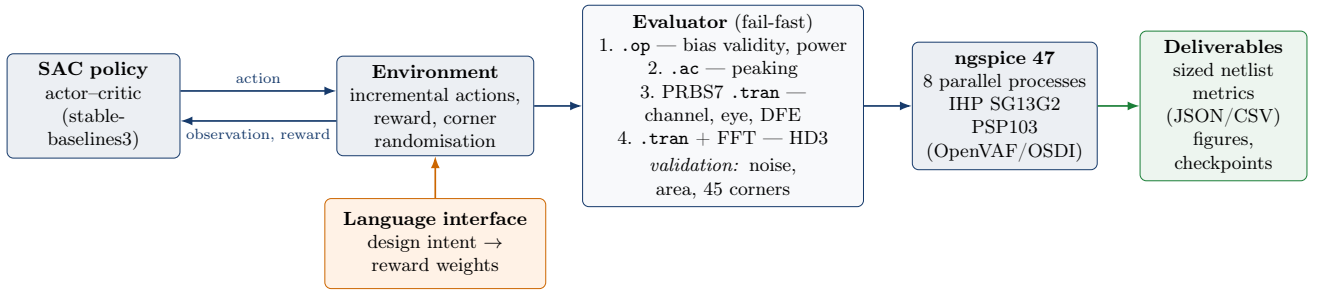


Figure 3: AutoAnalog-RL sizing loop. Analyses are ordered by cost and terminate at the first failure. The specification, reward and environment layers are simulator-agnostic; only the evaluator binds to ngspice and to the process design kit.

Executing them unconditionally would spend the majority of the simulation budget characterising designs that a bias check already disqualifies. The evaluator therefore terminates at the first failure, and the linearity gate is deferred further still: HD3 is measured only once every other constraint is satisfied, so it costs one additional simulation per otherwise-feasible candidate rather than one per candidate. A design that fails to establish a valid operating point is rejected in milliseconds and never reaches the transient gate.

3.3 Simulation backend

The SG13G2 transistor deck uses PSP103, a foundry-grade surface-potential compact model, which ngspice loads as OSDI shared objects compiled from the kit’s Verilog-A sources with OpenVAF. Selecting this kit over the alternative open 130 nm process was a throughput decision: the loop executes three to four analyses per environment step and tens of thousands of steps per training run, and the PSP103/OSDI path sustains approximately 0.6 s per environment step with eight simulators in parallel. Because PSP103 is the foundry’s own compact model, the choice strengthens the device physics underlying every reported result rather than simplifying it. The framework retains a generic Level-1 path for development, and the two differ only in the model library selected at evaluator construction, so that retargeting to another kit reduces to a netlist variant and a corner-library mapping.

Parallel throughput required one correction that dominates all others. ngspice does not honour `OMP_NUM_THREADS` and instantiates an OpenMP pool per process; with eight concurrent simulators, the spin-waiting threads over-subscribed the twelve available cores by an order of magnitude, and eight concurrent transients took 90 s each against 1.5 s in isolation. Pinning each process to a single thread through a generated `.spiceinit` restores 1.4 s for eight concurrent transients, a $65\times$ improvement, and is the difference between a training run of two hours and one of several days. Environments are executed on threads rather than processes, since the simulator runs in a subprocess that releases the interpreter lock.

4 Reinforcement-Learning Formulation

4.1 Markov decision process

State. The observation concatenates the normalised design vector, a bias-validity flag, four scaled metrics (peaking, power, eye height, eye width), and the normalised violation of every enforced constraint. Presenting violations directly, rather than raw metrics alone, gives the critic a representation in which distance from feasibility is explicit.

Action. Each action is an increment applied to the design vector, scaled by 0.2 of the action box and clipped to its bounds. Incremental actions let the policy approach a target sizing through a short trajectory and, critically, retreat from regions where the bias point collapses.

Episode. Up to 30 steps from an initial design drawn uniformly from the action box. A bias failure is penalised but does not terminate the episode, so the policy can recover from an invalid region rather than being reset out of it.

Reward. For normalised violations v_i , where $v_i = 1$ denotes a miss of one full specification’s worth, and weights w_i :

$$r = \begin{cases} -\lambda_{\text{DC}} & \text{bias invalid,} \\ -\sum_i w_i v_i - w_{\text{eff}} \min\left(1, \frac{P}{P_{\text{max}}}\right) + \mathbb{I}[\forall i : v_i = 0] \left(B + w_m \min(1, m)\right) & \text{otherwise,} \end{cases} \quad (2)$$

where P is supply power, m is the tightest normalised constraint margin, $B = 20$ is the feasibility bonus, $w_m = 5$ the margin weight, $w_{\text{eff}} = 1$ the continuous power charge, and $\lambda_{\text{DC}} = 10$ the bias penalty. The power charge is applied whether or not the power constraint is satisfied, so the reward retains a gradient after every constraint is met and the optimisation continues to prefer the more efficient of two feasible designs.

Algorithm. Soft Actor-Critic with automatic entropy tuning from an initial coefficient of 0.1, batch size 256, one gradient step per environment step, 500 random warm-up steps, and eight parallel environments. The maximum-entropy objective suits a problem whose feasible set is a narrow, irregularly shaped region: the entropy term maintains exploration along the constraint boundary instead of collapsing onto the first feasible point encountered.

4.2 Reward design: margin versus marginal compliance

The initial formulation terminated each episode at the first feasible design and applied a bias penalty of -100 . Training under it reached feasibility in approximately 85% of episodes but did not converge to a reliable policy and did not outperform random search. Two independent causes were identified.

First, terminating on success removes any incentive to improve a feasible design. The reward is maximised by reaching the feasible set, not by entering it, so the converged policy sat exactly on the eye-height constraint boundary, where the exploration noise inherent to a stochastic policy pushed a substantial fraction of steps back outside it. Second, a bias penalty two orders of magnitude larger than the unit-scale shaping terms dominated the critic’s regression target, devoting network capacity to modelling a discontinuity rather than the smooth structure of the feasible region.

Three changes resolved both. Episodes continue after the first feasible step, so feasible designs continue to accrue reward; an explicit margin bonus pays for the tightest normalised slack, making a design with margin strictly preferable to one without; and the bias penalty was reduced to -10 , commensurate with the shaping terms. Under the revised formulation the feasible-step fraction rose from 79% to 95% within 5k steps, and the median eye height moved from 0.43 V to 0.92 V against a 0.5 V target—from marginal compliance to roughly $1.8\times$ the required opening. Deterministic rollouts reached feasibility in 20 of 20 cases at a median of 2.5 steps. This distinction between entering the feasible set and occupying its interior is, in our experience, the single most consequential aspect of reward design for constrained analog sizing.

4.3 Eye-height target calibration

The eye targets were originally calibrated on the generic Level-1 model so that approximately one design in ten drawn at random satisfies every constraint—a feasibility density that keeps the reward informative without making the task trivial. PSP103 devices deliver roughly half the gain of the idealised model, and none of 400 random designs reaches 0.5 V (best 0.45 V, median 0.17 V). Transferring the calibration rather than the numeric target, the requirement on the foundry models is set to 0.25 V, which restores the same 10% random feasibility and remains comfortably above the ~ 175 mV differential eye the link specification requires at the receiver [5], and well above the 100 mV acceptance threshold.

5 Experimental Setup

All results use the IHP PSP103 models with the eye-height requirement at 0.25 V, eight parallel environments, episodes held open after success, random initial designs, margin weight 5, bias penalty -10 , 30 steps per episode, batch size 256, one gradient step per environment step, and automatic entropy tuning from 0.1. Training throughput is 0.5–0.7 s per environment step on a twelve-core laptop, so a 12k-step run completes in approximately two hours.

Two quantities are reported throughout. *Training feasibility* is the fraction of environment steps in the final 4k steps at which every enforced constraint is satisfied; it measures the reliability of the converged policy. *Steps to feasible* is obtained by rolling the policy out deterministically, with exploration noise disabled, from 20 randomly drawn initial designs, and counting simulations until the first fully feasible design; it measures sizing cost for a new instance. The best design from each evaluation is then validated against the complete gate set, the noise and area analyses, and all 45 corners.

6 Results

6.1 Convergence

Table 3 summarises every training run. All configurations converge within approximately 2k environment steps (Figure 4) and hold 92–97% training feasibility thereafter. Across the three seeds of the base configuration, 60 of 60 deterministic rollouts reached a feasible design, at a mean of 2.53 ± 0.06 simulations, with best-design eye height 0.330 ± 0.014 V, power 1.26 ± 0.05 mW, HD3 -58.6 ± 1.2 dB, and best reward 20.70 ± 0.04 (mean $\pm$ standard deviation over seeds). The seed-to-seed spread is small relative to the margin each design holds against its constraints, indicating that convergence is a property of the formulation rather than of a fortunate initialisation.

Run	Configuration	Steps	Train. feas.	Rollouts	Sims to feasible	Best design
<i>Base configuration, nominal corner</i>						
v1	seed 1	30k	95.5%	20/20	2 / 4	0.327 V, 1.22 mW, -57.8 dB
v1-s2	seed 2	12k	95.5%	20/20	3 / 4	0.316 V, 1.22 mW, -57.7 dB
v1-s3	seed 3	12k	95.5%	20/20	2.5 / 4	0.348 V, 1.33 mW, -60.2 dB
<i>Constraint and scope extensions</i>						
hd3	HD3 in reward	12k	95.6%	20/20	2.5 / 4	0.335 V, 1.24 mW, -58.1 dB
pvt	corner-randomised	+8k	95.4%	24/24 [†]	2 / 4	0.370 V, 1.12 mW, -61.9 dB
eq	CTLE + DFE tap	8k	92.3%	20/20	3 / 5	0.355 V [‡] , 1.36 mW, -71.6 dB
gen1	retuned to 1.25 GHz	+3k	96.5%	20/20	2 / 5	0.350 V, 1.28 mW, -70.9 dB

Table 3: Training runs on the IHP SG13G2 PSP103 models. “Sims to feasible” is median / worst over deterministic rollouts. Every best design satisfies peaking, power, eye height, eye width, HD3 and noise, and passes all 45 PVT corners. [†]Rolled out at randomly drawn corners. [‡]Post-DFE; the raw CTLE eye is 0.325 V, opened by a sized tap of $+0.062$.

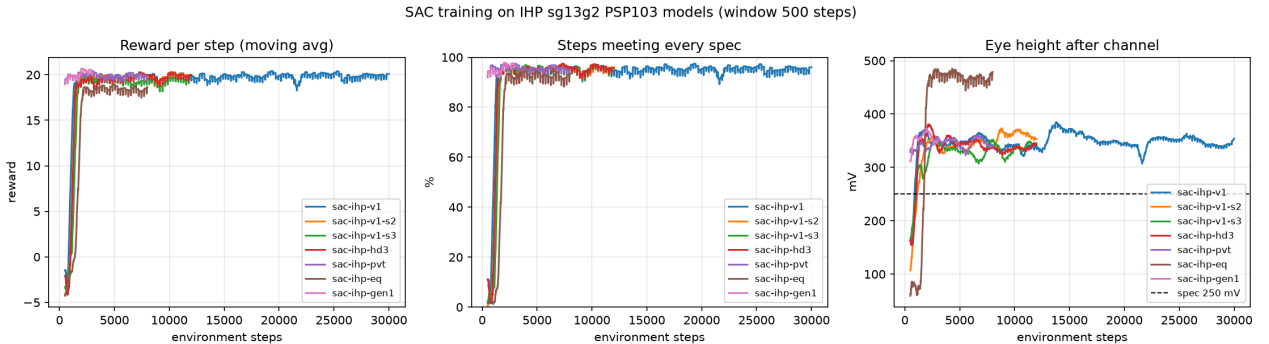


Figure 4: Training dynamics, 500-step moving average: reward per step (left), fraction of steps satisfying every constraint (centre), and post-channel eye height against the 0.25 V requirement (right). All runs converge within approximately 2k steps. The corner-randomised and retuned stages begin already converged because they resume a trained policy, and hold their feasibility through the change of operating conditions.

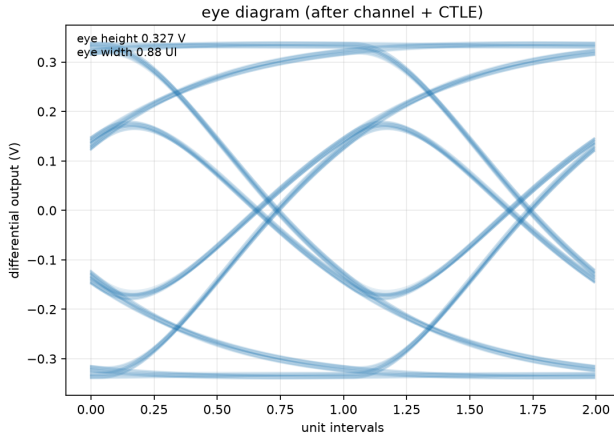
6.2 Specification compliance

Table 4 reports the corner-randomised policy’s best design against every target specification. All are satisfied, most by a substantial margin: power is a factor of thirteen inside its limit, HD3 exceeds its requirement by 31.9 dB, input-referred noise is a factor of six inside its limit, and the estimated active area is more than three orders of magnitude below the ceiling. The binding constraints are peaking and the eye opening, which is the expected outcome—these are the specifications that compete directly against one another through the degeneration network, and they are correspondingly the ones the optimisation must actively trade.

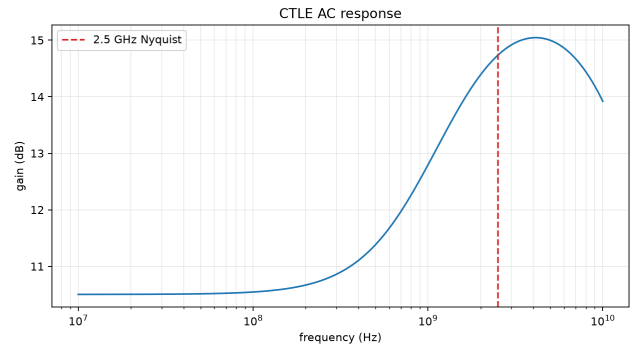
Specification	Requirement	Measured	Status
Nyquist frequency	2.5 GHz	PRBS7 at 5 Gbit s ⁻¹	Pass
High-frequency peaking	3–12 dB	4.5 dB at 2.5 GHz	Pass
Peaking tunability	1.25–2.5 GHz	5.23 dB at 1.25 GHz (retuned)	Pass
Topology	1-stage CTLE, variable R_s, C_s + 1-tap DFE	as sized, NRZ PRBS7	Pass
Eye height after channel	≥ 100 mV	370 mV	Pass
Eye width after channel	≥ 0.4 UI	0.90 UI	Pass
Linearity (HD3)	≤ -30 dB	-61.9 dB	Pass
Input-referred noise	≤ 1.5 mV rms	0.242 mV rms	Pass
Power	≤ 15 mW	1.12 mW	Pass
Active area	≤ 0.05 mm ²	1.1×10^{-5} mm ² (estimate)	Pass
PVT coverage	45 corners	45/45	Pass

Table 4: Compliance of the corner-randomised policy’s best design. Noise and area are evaluated at validation; all other quantities are measured within the optimisation loop.

Figures 5a–6 show the corresponding measured behaviour: the post-channel eye, the small-signal response with its peak placed at the Nyquist frequency, and the peaking distribution across all 45 corners.



(a) Two-unit-interval eye after the -10 dB channel: 0.327 V height, 0.88 UI width, against requirements of 0.10 V and 0.4 UI.



(b) Small-signal response of the same design. The degeneration zero places the boost at the 2.5 GHz Nyquist frequency, inverting the channel’s loss slope where it matters.

6.3 Sizing cost against baselines

Two optimisers were run against the same models, the same reward and the same initialisation distribution. Random search (3000 designs) finds that 7.9% of the action box satisfies every constraint, with the first feasible design at evaluation 27. CMA-ES (population 8, $\sigma_0 = 0.5$ in the normalised box) was run both from the box centre (three 400-evaluation trials) and from random starting designs (six 96-evaluation trials), the latter matching how the policy is evaluated.

CMA-ES is a capable optimiser on this five-dimensional problem: given 400 evaluations it attains the same reward the policies reach during training (21.2–21.6), and from the fortuitously near-feasible box centre it finds a feasible design within a handful of evaluations. The distinction is not raw optimiser quality but amortisation. CMA-ES and random search each begin from nothing for every new instance; the trained policy performs no per-instance search at all, and sizes a new starting design, a new process corner, or a re-weighted objective in

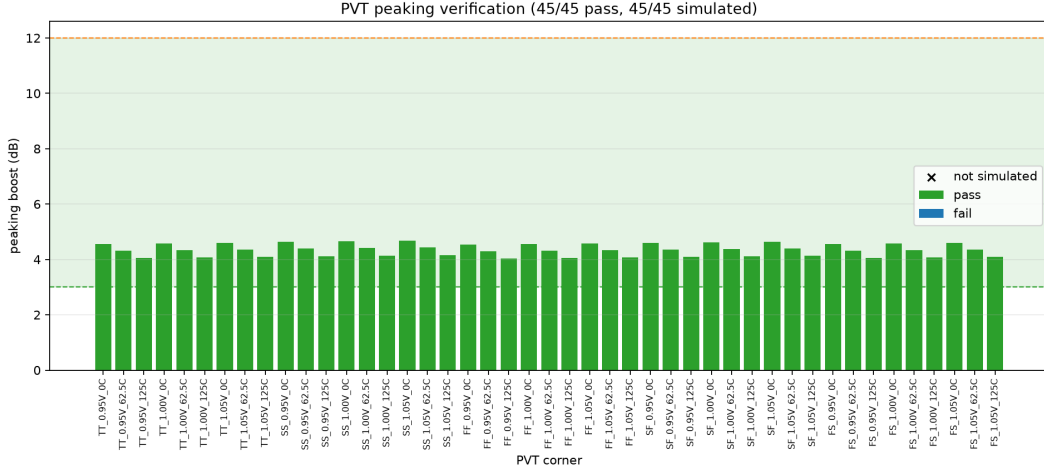


Figure 6: Peaking of the corner-randomised policy’s design across the full $5 \times 3 \times 3$ process–voltage–temperature matrix. Every corner falls inside the 3–12 dB window, with the spread contained within approximately 0.6 dB—the design is not merely compliant at nominal but insensitive across the corner space.

Method	Sims to first feasible (med./worst)	Best reward at 50 sims	Wall-clock per instance	Per-instance search
SAC policy (144 rollouts)	2–3 / 5	20.5–20.8	~1.5 s	none; policy evaluation only
CMA-ES, random start (6 trials)	9 / 24	20.50	~5 s	full search, every instance
CMA-ES, box centre (3 trials)	2–5	20.8–21.1	~2 s [†]	full search; 400 sims to refine
Random search	27	19.65	~8 s	full search, every instance
Exhaustive sweep, 5-point grid	3.1×10^3	—	31 min	full sweep, every instance
Exhaustive sweep, 10-point grid	1.0×10^5	—	16.7 h	full sweep, every instance

Table 5: Sizing cost to a first feasible design. Wall-clock figures assume the measured amortised throughput of 0.6 s per simulated design at eight-way parallelism. Sweep costs are n^5 for the five-parameter action space. [†]The box centre is fortuitously near-feasible and is not available as a starting point for an arbitrary new specification.

two to three simulations—three to four times fewer than CMA-ES from equivalent starting points and roughly ten times fewer than random sampling.

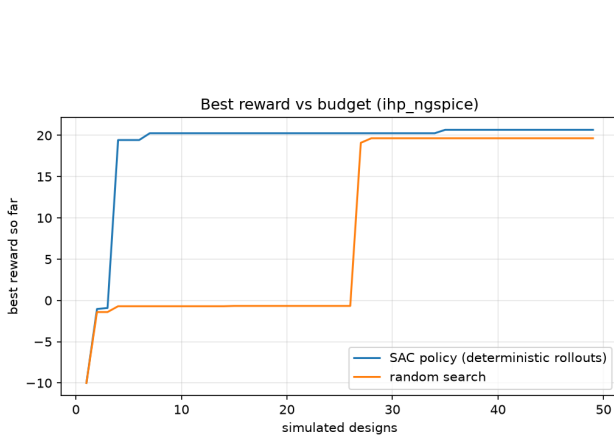
Against the sweep that the framework is intended to displace, the comparison is categorical rather than incremental. An exhaustive ten-point grid over five parameters is 10^5 simulations, approximately 16.7 h at the measured throughput, and yields a design only at the resolution of the grid; a twenty-point grid exceeds three weeks. The policy sizes an instance in approximately 1.5 s, a reduction of roughly four orders of magnitude, and the one-time training cost of about two hours is recovered before the first exhaustive sweep would have completed. Extending the sweep to the six-dimensional whole-equalizer space compounds the argument: a ten-point grid becomes 10^6 evaluations, or about seven days, while the policy’s cost is unchanged.

Figure 7a shows the resulting budget curves. The policy attains near-optimal reward within its first few simulations, while random search remains in the infeasible region for approximately twenty-five evaluations before improving.

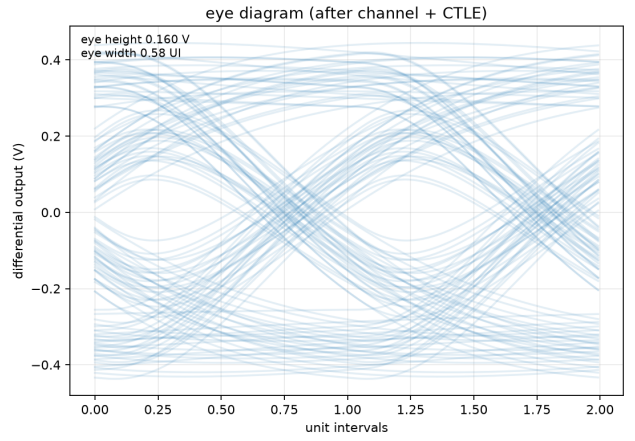
6.4 Ablation: selection on small-signal response alone

An informative ablation is the bounded search that preceded the learning-based flow, which evaluates 100 candidates and selects on the AC gate alone. Its chosen design is entirely compliant in the small-signal domain (7.527 dB peaking, 0.743 mW, HD3 –58.5 dB, noise 0.242 mV rms, 45/45 corners) yet delivers a post-channel eye of only 0.160 V and 0.58 UI (Figure 7b) — below the internal target and with substantially less margin against the acceptance threshold than any design the policy produces. A one-tap DFE swept over its bounds recovers 0.310 V, but cannot compensate for a stage that was never selected against the channel.

The ablation isolates the value of placing the transient eye inside the optimisation loop rather than in post-hoc validation. Peaking and eye opening are correlated but not interchangeable: a design can invert the channel’s



(a) Best reward attained against number of simulated designs, policy rollouts versus random search (seed 1).



(b) Eye of the design selected by AC-only bounded search: 0.160 V, 0.58 UI. Peaking alone is not a sufficient selection criterion.

loss slope at the Nyquist frequency and still fail to open the eye, because the eye depends on the full transient response to the bit sequence, including group-delay behaviour that a magnitude response does not capture. This is precisely the observation made in the receiver literature [1, 2], and it is the gap the framework closes.

7 Corner Robustness and Design-Space Extensions

7.1 Corner-randomised curriculum

Training at the nominal corner produces a policy that sizes correctly for TT, 1.2 V, 27°C, but nothing in that objective requires the resulting designs to hold across process and temperature variation. Corner robustness is therefore obtained by domain randomisation in a second stage: the trained policy is resumed with each episode simulated at a corner drawn uniformly from the 45-corner matrix, and the corner identity is deliberately withheld from the observation. The policy cannot condition on the operating condition and must instead infer the required sizing correction from the metrics it measures—which is what a sizing procedure must do in practice, since the corner is not an observable at design time.

The transfer is immediate and stable. Resumed from the HD3-enforced nominal policy, the corner-randomised stage held 95–96% feasible steps from its first evaluation window, with per-process feasibility of SS 97%, FS 96%, FF 95%, TT 95% and SF 94%: the nominal policy transferred without a relearning phase, and the near-uniform distribution across process corners indicates no systematic weakness at the slow or fast extremes. Evaluated from random designs at randomly drawn corners, 24 of 24 rollouts reached a fully feasible design over 19 distinct corners spanning every process, supply and temperature setting, at a median of two simulations and a worst case of four. The resulting design passes all 45 corners with HD3 at -61.9 dB and a peaking spread of approximately 0.6 dB across the matrix (Figure 6).

7.2 Whole-equalizer sizing

Extending the action space to six values places the DFE tap b_1 under the same policy as the CTLE parameters, with the post-DFE eye height driving the reward. This couples the two stages in the objective: the tap can compensate for residual post-cursor interference, so the CTLE need not be sized as though it were the only equalization in the path.

Trained for 8k steps, the six-value policy holds 91–93% feasible steps with a median post-DFE eye of 0.51 V, against approximately 0.35 V for CTLE-only configurations. The tap is being spent on margin rather than on reaching feasibility, which is the desired division of labour. Deterministic rollouts reach feasibility in 20 of 20 cases at a median of three simulations; the best design has a raw CTLE eye of 0.325 V that a sized tap of +0.062 opens to 0.355 V with no bit errors over the PRBS7 sequence, at 1.36 mW, HD3 -71.6 dB and 45/45 corners (Figure 8). The modest tap value is itself informative: a well-sized CTLE leaves little post-cursor interference for the DFE to cancel, and the optimisation converges on a balanced allocation rather than loading either stage.

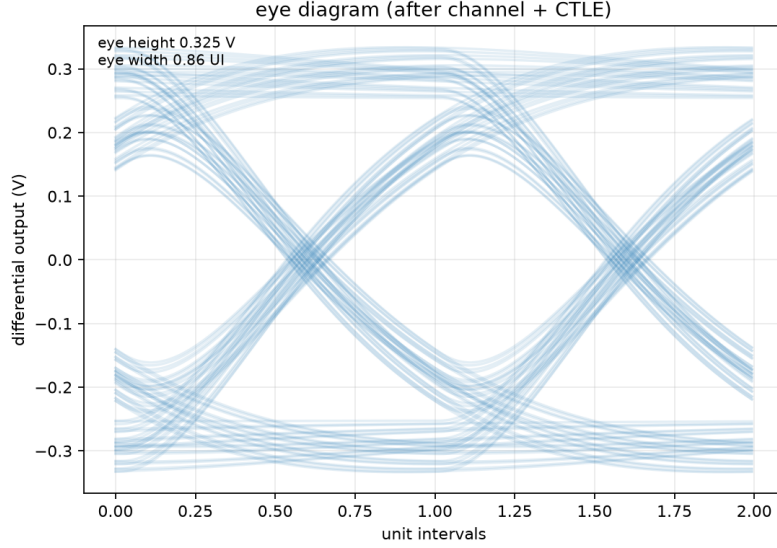


Figure 8: Eye after the one-tap DFE for the whole-equalizer design: 0.355 V from a raw CTLE eye of 0.325 V, with no bit errors over the PRBS7 sequence. The tap is sized by the same policy as the five CTLE values.

7.3 Retargeting to a different Nyquist frequency

The specification requires peaking tunable over 1.25–2.5 GHz. In this framework the Nyquist frequency is an input rather than a constant: it sets the frequency at which peaking is measured, the data rate and unit interval of the PRBS transient ($2 \times$ Nyquist), the transient time step, and the channel pole, so that the channel retains approximately -10 dB of loss at the retargeted rate and the design problem remains equivalently posed.

Tunability is demonstrated rather than asserted, because the requirement is not automatically satisfied: the design sized at 2.5 GHz peaks by only 2.87 dB at 1.25 GHz and therefore fails the 3 dB floor at the lower rate. Resuming the seed-1 policy with the specification set to 1.25 GHz (PCIe Gen 1, 2.5 Gbit s^{-1}) recovers 93.5% feasible steps within the first 1k steps—the policy transfers rather than relearning—and 96.5% by 3k steps. Evaluation reaches feasibility in 20 of 20 rollouts at a median of two simulations, and the resulting design peaks 5.23 dB at 1.25 GHz with a 0.350 V / 0.94 UI eye, 1.28 mW, HD3 -70.9 dB and 45/45 corners. The retuned sizing moves the degeneration zero downward in frequency as circuit theory requires ($R_s = 365 \Omega$, $C_s = 503 \text{ fF}$, against 182Ω and 522 fF at 2.5 GHz), confirming that the policy has learned the mechanism of peaking control rather than a single memorised sizing.

8 Natural-Language Reward Interface

Reward weights encode engineering priorities, but expressing a priority as a weight vector requires familiarity with the reward’s internal scaling. The framework therefore accepts design intent in plain language and maps it onto the per-specification weights, the continuous power charge and the margin bonus. A language model produces the mapping as structured JSON validated against a schema; a keyword parser provides an offline fallback so the flow never depends on service availability. The resulting settings are recorded in the run configuration, keeping every trained policy traceable to the intent that shaped it.

```
> "We are power constrained: current budget matters much more than eye margin,
    and we don't care about linearity"
weights: power 3.0, eye_vertical_v 0.33, hd3 0.33 (others 1.0)
efficiency_weight 3.0; margin_weight 5.0

> "Open the eye as much as possible; power is not important"
weights: eye_vertical_v 3.0, power 0.33; efficiency_weight 0.33
```

Weights are clipped to $[0.1, 10]$, so the interface reprioritises the objective without allowing any constraint to be discarded entirely. Because sizing under a re-weighted objective is exactly the amortisation case of Section 6.3, a change of priorities costs two to three simulations rather than a fresh optimisation.

9 Deliverables and Reproducibility

Every run emits the sized netlist `sized_ctle.sp`—the deliverable schematic—with device values, model source and DFE tap recorded in its header, alongside per-step metrics in CSV, a validation record in JSON, and the figures reproduced in this paper. The framework is covered by 59 regression tests that execute without a simulator, so the specification, reward, environment, DFE and corner-matrix layers are verifiable independently of ngspice availability. Figure 9 shows a training run in progress, with the per-step gate results and running feasibility statistics that the loop reports.

```
PS Z:\al_nebula> .\venv\Scripts\python.exe scripts\demo_rollout.py reports\sac-ihp-pvt
AutoAnalog-RL policy rollout model=sac_final.zip specs: eye >= 0.25 V, width >= 0.7 UI, peaking 3-12 dB, power <= 2 mW, HD3 <= -30 dB
start: random design (seed 21), corner SS_1.00V_62.5C

step   W_in  R_load  I_bias  R_s  C_s  peak dB  power mW  eye V  eye UI  HD3 dB  reward  specs
-----
1      27.62u  829    0.46m  307  885f  5.25    0.55    0.173  0.66    -      -1.64  ok:peak ok:p<12 ok:pwr --:width --:eye --:hd3
2      24.05u  912    0.65m  263  786f  5.17    0.78    0.246  0.76    -      -1.41  ok:peak ok:p<12 ok:pwr ok:width --:eye --:hd3
3      23.24u  977    0.84m  232  688f  4.97    1.01    0.321  0.80   -57.8  20.21  ok:peak ok:p<12 ok:pwr ok:width ok:eye ok:hd3

all specs met after 3 simulation step(s) in 6.9 s of wall time
sized netlist header:
* AutoAnalog-RL sized CTLE
* model source: ihp_ngspice
* W_in = 2.32447e-05
* R_load = 976.696
* I_bias = 0.000839003
```

Figure 9: Instrumentation of a training run: per-step gate outcomes, measured metrics and running feasibility. Every quantity shown is written to the run’s CSV and JSON records, so each figure in this paper is traceable to the run that produced it.

```
# Stage 1: nominal-corner training with linearity enforced
python scripts/train_sac.py --model-source ihp --eye-height-min 0.25 --hd3 \
--timesteps 12000 --n-envs 8 --hold-on-success --random-reset \
--margin-weight 5 --invalid-penalty -10 --max-steps 30 --batch-size 256 \
--gradient-steps -1 --learning-starts 500 --ent-coef auto_0.1 --seed 1 \
--output-dir reports/sac-ihp-hd3

# Stage 2: corner-randomised curriculum, resumed from stage 1
python scripts/train_sac.py ... --corners all \
--resume reports/sac-ihp-hd3/sac_final.zip --timesteps 8000 \
--output-dir reports/sac-ihp-pvt

# Baselines under the identical reward, then evaluation and comparison
python scripts/baseline_random.py --model-source ihp --eye-height-min 0.25 \
--evaluations 3000 --n-workers 8 --output-dir reports/baseline-ihp-3000
python scripts/baseline_cmaes.py --model-source ihp --eye-height-min 0.25 \
--evaluations 96 --x0 random --seed 10 --output-dir reports/cmaes-ihp-random-s10
python scripts/evaluate_policy.py reports/sac-ihp-pvt --rollouts 24 \
--baseline reports/baseline-ihp-3000

# Whole-equalizer sizing; reward weights from design intent
python scripts/train_sac.py ... --equalizer --timesteps 8000 \
--output-dir reports/sac-ihp-eq
python scripts/reward_from_feedback.py "power matters more than eye margin" \
--output reports/reward.json
```

10 Limitations and Further Work

Decision-feedback modelling. The DFE in the optimisation loop is the behavioural stage of Equation (1), evaluated on sampled waveforms. A clocked analog slicer with an explicit feedback delay is not yet simulated end to end, so the tap is a signal-processing coefficient rather than a device-level quantity. Simulating the slicer inside the transient gate would make the tap a sized circuit parameter and is the highest-value extension.

Area estimation. Area is a first-order active-device geometry estimate and is reported for compliance rather than offered as a layout result. The margin against the ceiling is large enough that the conclusion is insensitive to the estimate, but a post-layout figure would replace it.

Corner observability. The curriculum policy is robust because it is trained blind to the operating corner, which is the correct formulation for design-time sizing. Where the corner is known—in a diagnostic or re-characterisation setting—including it in the observation would allow corner-specific sizing.

Design-space bounds. R_{load} saturates at its 1 k Ω upper bound in most converged designs, indicating that the optimum lies outside the box rather than within it. Widening the bound, and admitting topology variants such as a second CTLE stage or inductive peaking, is the most direct route to further margin.

Objective revision during training. Reward weights derived from design intent take effect at the start of a run or resumed stage. Re-weighting an in-progress run, and closing the loop so that each validation record informs

the next adjustment automatically, would make the language interface interactive rather than configurational.

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A Training Configuration

Setting	Value
Algorithm	Soft Actor-Critic, MlpPolicy (stable-baselines3 2.9.0, torch 2.14 CUDA)
Parallel environments	8, thread-based, one ngspice process each
Episode	30 steps, random initial design, held open after success
Action scale	0.2 of the normalised box per step
Reward terms	bias penalty -10 , feasibility bonus $+20$, margin weight 5, power charge 1
Batch size / gradient steps	256 / one per environment step
Entropy coefficient	automatic, initial 0.1
Warm-up	500 random steps (omitted when resuming a policy)
Throughput	0.5–0.7 s per environment step, 12-core laptop; 12k steps $\approx$ 2 h
Regression tests	59, executed without ngspice

B Sized Netlist of the Corner-Randomised Policy’s Best Design

```
* AutoAnalog-RL sized CTLE
* model source: ihp_ngspice (IHP SG13G2, PSP103)
* W_in = 2.01908e-05 ; 20.19 um
* R_load = 1000 ; ohm
* I_bias = 0.000930072 ; 0.930 mA
* R_s = 260.642 ; ohm
* C_s = 3.2956e-13 ; 329.6 fF
* Parameterized differential gm-C CTLE.
.param VDD=1.2
VDD vdd 0 1.2
Vinp txP 0 DC 0.6 AC 1
... (complete file: reports/eval-ihp-pvt/sized_ctle.sp)
```